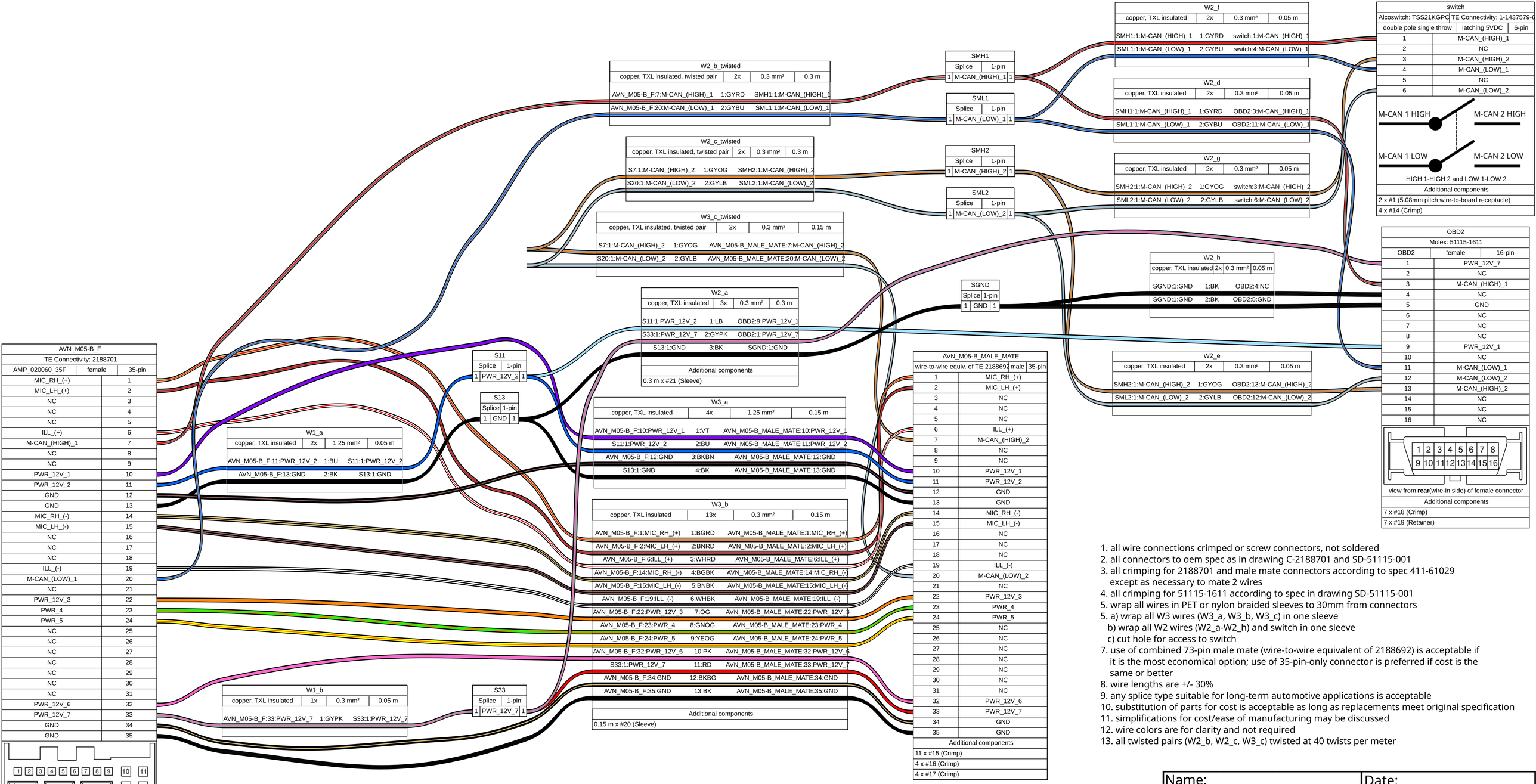

1. all wire connections crimped or screw connectors, not soldered
2. all connectors to oem spec as in drawing C-2188701 and SD-51115-001
3. all crimping for 2188701 and male mate connectors according to spec 411-61029 except as necessary to mate 2 wires
4. all crimping for 51115-1611 according to spec in drawing SD-51115-001
5. wrap all wires in PET or nylon braided sleeves to 30mm from connectors
5. a) wrap all W3 wires (W3_a, W3_b, W3_c) in one sleeve
   b) wrap all W2 wires (W2_a-W2_h) and switch in one sleeve
   c) cut hole for access to switch
7. use of combined 73-pin male mate (wire-to-wire equivalent of 2188692) is acceptable if it is the most economical option; use of 35-pin-only connector is preferred if cost is the same or better
8. wire lengths are +/- 30%
9. any splice type suitable for long-term automotive applications is acceptable
10. substitution of parts for cost is acceptable as long as replacements meet original specification
11. simplifications for cost/ease of manufacturing may be discussed
12. wire colors are for clarity and not required
13. all twisted pairs (W2_b, W2_c, W3_c) twisted at 40 twists per meter

Name: Hyundai/Kia M-CAN to OBD dongle	Date: 2026-03-17
Drawing Number: HKMOBD_SR_ES-003	Drawn by: Tyler Harvey